Wave Release Coordination under Vertical Resource Constraints in Multi-Storey AMR Warehouses

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Abstract

Multi-storey warehouses staffed by floor-bound autonomous mobile robots (AMRs) couple wave-release decisions to a small number of freight elevators whose per-trip capacity is the binding constraint. We formalise this as the **wave release coordination problem under vertical resource constraints**: a two-stage scheduler that selects wave composition tactically and dispatches AMR–elevator execution operationally. We contribute: (i) a three-dimensional structured decomposition $\Phi = (C, I, T)$ with physically interpretable per-unit coefficients; (ii) a **Bound-and-Gap framework** whose decomposition theorem splits the value of wave-structure information into a partition-intrinsic upper bound and a Φ-policy lower bound (mean GAP 5.83 %, CI excludes zero in 6/6 cells); (iii) a **Model-Dominance Hedge Rule** collapsing under per-wave stochastic dominance ($\mathbb{P}[M_2 \geq M_1] =$ 99.4 % across $c \in \{2, 3, 4, 5\}$) to a closed-form rule with worst-case loss 3.1–9.9 %. Across 18 000 regime-sweep simulations, the predictive value of structured features grows ten-fold with elevator-bottleneck relief. A pre-registered 14 400-simulation policy test produces a **tactical-operational substitutability map**: destination-clustered batching cuts mean makespan up to 9 % exactly in the cells our framework flags prospectively (6/6 cells correctly classified). Robustness across $\sim$170 000 total simulations covers building heights $F \in \{5, 7, 9\}$, heterogeneous elevator pools, and service-time CV up to 131 %.

1 Introduction

1.1 Motivating scenario

Consider a multi-storey fulfilment warehouse with F floors, E freight elevators of per-trip capacity c, and a fleet of floor-bound AMRs. An order travels horizontally on its source floor, vertically by elevator, and horizontally on its destination floor; the elevator is the only vertical channel and is shared by the entire fleet. Operations managers control two layers. The **operational** layer assigns AMRs to orders and sequences elevator boardings — a problem with substantial existing literature, from order batching in vertical lift modules to SAT-based optimal robot–lift schedules. The **tactical** layer chooses *which orders co-occupy a release window* — what we call a wave. Wave composition determines how many vertical transitions the elevators must serve, in which directions, and with what temporal stagger; it therefore sets the difficulty of the operational problem before any robot moves. Despite this, wave composition has not been treated as a decision variable jointly with elevator capacity in the AMR-warehouse literature. The omission matters: the wave layer is the cheapest place to relieve a vertical bottleneck, because it costs only a re-batching of already-released orders, with no change to floor layout or fleet size.

1.2 Why this is hard now

Three threads of recent work each touch part of this problem and stop short of the wave–elevator coupling.

Order batching for **vertical lift modules** (Lenoble et al., 2018) optimises pick rounds inside a single device; the relevant decision is which items to retrieve in one shuttle trip, not which orders to release into a building-scale wave. The mathematics is local to the device.

Multi-agent **robot–lift scheduling** (Chakravarty et al., 2025) takes a fixed task set and computes provably optimal lift schedules via Boolean satisfiability. Lift availability appears as a constraint, not as a design lever for the upstream wave layer; task selection is exogenous.

Tier-captive shuttle systems (Wu et al., 2024) and multi-level RMFS architectures (Keung, 2023) study performance with floor-attached transport (one shuttle per tier), where vertical coupling is structurally weaker. The closer setting — flexible AMRs that can occupy any floor and queue at any elevator — produces a different bottleneck topology that these analyses do not address.

Two further methodological threads bear on our framing. **Robust scheduling under model uncertainty** (Wiesemann et al., 2014; Lu & Shen, 2021) handles parametric uncertainty in known model classes, whereas the elevator literature offers genuinely *different* abstractions (throughput aggregation versus true co-occupancy batching) whose disagreement is structural, not parametric. **Prediction-to-decision regret** (Elmachtoub & Grigas, 2022; Vera et al., 2022) bounds algorithmic loss when a learned predictor drives a downstream optimiser; we are interested in a complementary quantity — the *information gap* between an oracle on the same feature partition and a Φ-informed policy — that those bounds do not name.

1.3 Two empirical surprises that drive the paper

Two specific findings from a 14 400-simulation sweep across six (regime, elevator-model) cells motivate the methodological apparatus.

Surprise A — wave structure has bounded but consistent value. A Φ-informed corner pick beats a random wave in 6/6 cells, but the lift is only 1.3–7.4 %, while the *oracle* corner — the best of the four quartile bins on (C, I) — wins 5–15 %. Neither zero nor full: the gap between informed policy and oracle is itself an object that needs a name and a decomposition. We provide both (§ 3.1): a Bound-and-Gap framework whose decomposition theorem

$$\mathrm{GAP} \;=\; H_{\mathrm{up}} \;+\; M_{\Phi}$$

separates partition-intrinsic difficulty from Φ-policy slack, and a partition-refinement corollary that explains, rather than rescues, the inflation of upper bounds under finer feature grids.

Surprise B — elevator-modelling assumption flips a structural sign. Re-running identical waves under throughput aggregation versus true co-occupancy batching flips the sign of the entropy coefficient $\beta(C)$ in 2 of 3 c=2 regimes. Bootstrap stability is 76–85 % within regime, so we treat the sign-flip as a coarse diagnostic, not a fine classifier; the load-bearing fact for our hedge rule is the underlying *per-wave dominance*: true-batching makespan exceeds throughput-abstraction makespan in 92.5–100 % of waves across all 15 cells. From this, the minimax wave-corner selection (§ 3.2) collapses in $c = 2$ regimes to a one-line rule — *always pick the corner the true-batching model prefers* — with worst-case loss 3.1–9.9 %.

1.4 Contributions and roadmap

We make three contributions.

(C1) Problem formulation. We formalise the wave release coordination problem under vertical resource constraints — a two-stage scheduler coupling tactical wave composition and operational AMR–elevator execution under a hard per-trip capacity. Related precedents address batching at the device level (Lenoble et al., 2018), optimal scheduling for fixed task sets (Chakravarty et al., 2025), or tier-captive transport (Wu et al., 2024; Keung, 2023); none treat wave composition jointly with building-scale vertical coupling and flexible fleet routing.

(C2) Methodology — three components. **(C2-Φ)** A three-dimensional structured decomposition $\Phi = (C, I, T)$ — vertical concentration, directional imbalance, temporal clustering — with physically interpretable per-unit coefficients. (A methodological caveat: throughput-abstraction and true-batching regressions agree on wave rank order, $r \geq 0.85$, but disagree on $\beta(C)$ sign, *motivating* the hedge rule below.) **(C2-M4)** A **Bound-and-Gap framework** with a decomposition theorem GAP $= H_{\text{up}} + M_{\Phi}$ (non-negative components, partition-refinement monotonicity), converting the gap into a per-cell diagnostic. **(C2-M5)** A **Model-Dominance Hedge Rule**: under per-wave stochastic dominance, the minimax corner collapses to "follow M_2" with worst-case loss 3.1–9.9 %, and a one-sided ε-bound predicts when the collapse remains sharp. Two confirmations: M3 stochastic experiment exhibits the predicted knife-edge at E2_c2 $\sigma = 0.20$, and a directional-elevator extension ($\mathbb{P}[M_4 \geq M_2] = 97.7\,\%$) has argmin stability tracking the batching axis exactly as Corollary M5.2 predicts. **M4 and M5 share the same applicability domain** (operationally active batching) — the Bound-and-Gap framework is prospective precisely where the Hedge Rule is sharp.

(C3) Insights — three findings.

(i) H_{up} varies ten-fold with elevator-bottleneck relief and, **within M5's applicability domain, prospectively classifies all 6 cells** on whether destination-clustered batching will beat FIFO.

(ii) $\mathbb{P}[M_2 \geq M_1] = 99.4\,\%$ across $c \in \{2, 3, 4, 5\}$; the Hedge Rule generalises beyond the $c = 2$ case it was derived for.

(iii) A **tactical-operational substitutability map**: non-substitutability concentrates in the elevator-lever-dominated, tight-wave regime and decays along both the capacity-slack and wave-looseness axes.

The remainder of the paper develops C1 (§ 2), C2 (§ 3), and C3 (§ 4), discusses managerial implications (§ 7), states limitations (§ 8), and concludes (§ 9).

2 Problem formulation

[draft section to follow]

3 Methodology

3.1 Bound-and-Gap framework (C2-M4)

[draft section to follow]

3.2 Model-Dominance Hedge Rule (C2-M5)

[draft section to follow]

4 Results

[draft section to follow]

References

Lenoble, N., Frein, Y., Hammami, R. (2018). Order batching in an automated warehouse with several vertical lift modules: Optimization and experiments with real data. *EJOR*.

Chakravarty, A., et al. (2025). Toward Optimal Multi-Agent Robot and Lift Schedules via Boolean Satisfiability. *Mathematics*, MDPI.

Wu, J., et al. (2024). Research on Inbound Jobs' Scheduling in Four-Way-Shuttle-Based Storage System. *Processes*, MDPI.

Keung, K. L. (2023). Multi-level RMFS architecture and analytics. *FSMJ*.

Wiesemann, W., Kuhn, D., Rustem, B. (2014). Robust Markov decision processes. *Mathematics of Operations Research*, 38(1), 153–183.

Lu, M., Shen, Z.-J. M. (2021). A review of robust operations management under model uncertainty. *POMS*, 30(6), 1927–1943.

Elmachtoub, A. N., Grigas, P. (2022). Smart "Predict, then Optimize". *Management Science*, 68(1), 9–26.

Vera, A., Banerjee, S., Gurvich, I. (2022). Online allocation and pricing: Constant regret via bellman inequalities. *Operations Research*.